Resolution Geometry: Operational Agreement and Geometric Descent

What the Agreement of Many Observers Establishes, What It Does Not, and How Their Records Assemble into One Geometry

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Abstract

RG is not a theory of how observers are made to agree. It is a theory of what their agreement establishes, what it fails to establish, and how compatible records assemble into geometry. We state this as the *Operational Agreement and Geometric Descent Principle*: six conditions under which several observers may be said to have reconstructed the *same* geometry. We prove it from the gluing (sheaf) lemma and Čech descent.

We then test it in seven executed series, with every random draw seeded.

A. Agreement versus closure. Three observers of a Penrose staircase agree pairwise on every overlap (residual ≤ 0.016). The loop holonomy is $|\delta| = 0.309 \pm 0.003$ against $2\pi\kappa = 0.314$. RG rejects the false global height map, keeps every local reading, and types the obstruction. A miscalibrated conversion among four observers is located from triangle residuals.

B. Different lenses. Ambient chords in metres, coarse geodesics in other units, and a two-chart stereographic atlas with its metric in centimetres are applied to one sphere. All three give the Euler characteristic 2 exactly, and curvature $\times\,\mathrm{rod}^2$ equal to $0.0330 \pm 0.5\%$, while their raw readings differ by a factor of 100. A one-dimensional loop lens cannot see curvature without a declaration.

C. Forgetting. Each observer masks a different cap and sees about 91%, yet jointly they see 100%. A cap hidden from all is a fibre: an alternative geometry inside it leaves every record unchanged (change outside: 0).

D. Orientation by redundancy. Relative orientations are recovered, with error falling from 0.30 to 0.13 to 0.058 as the degree rises. RG abstains when the spectral gap is too small. The global rotation is null (error 2.31) until one anchor fixes it (0.14). A corrupted observer is located.

E. Scale null. Observers at scales 0.1, 1 and 10 agree on ratios. Absolute size is null until a rod anchors it, after which the anchored lengths agree to 0.6%.

F. No faked structure. Shuffled labels, pure noise and a reflected chart are accepted in 0/200 trials each; genuine overlaps in 200/200.

G. Edges and sampling. The transition uncertainty scales as $m^{-1/2}$. A single overlap sample cannot certify a transition, so RG abstains. Covers must be good.

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1 The question

When several observers agree, what have they established? Agreement on every pairwise overlap is evidence of something. It is not, in general, evidence of a single global geometry. The difference is exactly the difference between local compatibility and global descent. RG's print already contains the tools: quotients and descent (L0–L19), rooms and the obstruction tower (L129–L131), atlas gluing (A6), mesh closure under coverage, unit anchors (L94–L109), and the typed nulls. This paper assembles them into one principle and tests it.

2 The principle

Principle 2.1 (Operational Agreement and Geometric Descent)**.** A common geometry is justified on a declared domain when:

(1) each protocol p_i defines a reproducible observation law on its region U_i;

(2) the regions lie in a declared common physical domain $X = \bigcup U_i$;

(3) each overlap conversion $g_{ij} : U_i \cap U_j \to G$ preserves the declared shared observables;

(4) the conversions satisfy the cocycle condition $g_{ij}g_{jk}g_{ki} = 1$ on every nonempty triple overlap;

(5) the reconstructed structure descends: it is independent of the protocol representation;

(6) unresolved global alternatives, such as nontrivial Čech classes, hidden fibres and nulls, are retained rather than silently identified.

Theorem 2.2 (Descent of geometry)**.** *Assume (1)–(5), with gauge group G acting on local reconstructions s_i that satisfy $s_j = g_{ij} \cdot s_i$ on overlaps.*

(a) There is a global object S (a G-torsor, i.e. a twisted global geometry) whose restriction to U_i is s_i. It is unique up to a global gauge, which is the null of the declared theory.

(b) S is a global chart*, i.e. trivializable, iff the Čech class $[g] \in H^1(\{U_i\}; G)$ vanishes. Otherwise S exists only as a twisted object, a room or obstruction in RG's typing, and every global chart is false.*

(c) If (4) fails on a nonempty triple overlap, no *global object exists. Some conversion is wrong, and the data contradict the claimed protocol model.*

Proof. (a) This is the gluing lemma for sheaves of local sections with transition functions satisfying the cocycle condition: the clutching construction. Uniqueness up to a global gauge follows because two gluings differ by a G-valued function compatible with all g_{ij}, which is constant on connected X for discrete G, and a global gauge in general.

(b) A global section exists iff the cocycle is a coboundary, $g_{ij} = h_j h_i^{-1}$.

(c) On $U_i \cap U_j \cap U_k \neq \emptyset$, the relations $s_j = g_{ij} s_i$, $s_k = g_{jk} s_j$ and $s_i = g_{ki} s_k$ force $g_{ij} g_{jk} g_{ki} = 1$ wherever the stabilizer of s_i is trivial. A violation is therefore inconsistent with *any* assignment of local sections.

For the imported framework see [1, 2]; for its RG placement see L131 and Structural Closure. □

Corollary 2.3 (Pairwise agreement is not global closure)**.** *On a cover whose triple overlaps are empty (three arcs on a circle), every pairwise agreement can hold while the class $[g] = \delta$ is nonzero. In that case the observers have reconstructed a twisted geometry, not a global chart.*

Decision rule. With estimated conversions and their uncertainty σ_δ, RG decides as follows:

- accept a global chart if $|\hat{\delta}| < 3\sigma_\delta$;
- reject it, typing the obstruction, if $|\hat{\delta}| > 5\sigma_\delta$;
- abstain in between.

These thresholds are declared, not derived.

3 Series A: agreement versus closure

A1, Penrose staircase. Three observers cover a circle with arcs $[0, 2.4]$, $[2.0, 4.5]$ and $[4.1, 2\pi + 0.3]$. Each reads a local height $\kappa\phi + c_i$ with an unknown zero c_i and noise 0.01, with 40 samples per overlap.

κ	$\lvert\hat{\delta}\rvert$	$2\pi\kappa$	σ_δ	verdict
0	0.0004	0	0.0038	global map accepted
0.002	0.0122	0.0126	0.0038	abstain (3.2σ)
0.05	0.309	0.314	0.0032	rejected: holonomy
0.2	1.251	1.257	0.0037	rejected: holonomy

Every pairwise overlap agrees, with residual sd between 0.012 and 0.016. A naive pipeline that glues pair by pair builds a height map with a seam of size $|\delta|$. RG instead keeps all local readings, rejects the global height map, and reports the correct global object: a $\mathbb{Z}$-cover (helix) with holonomy δ. The sign of $\hat{\delta}$ records the loop's orientation.

A2, a miscalibrated conversion. Four observers share a common region, and the conversion on edge 1–3 is wrong by 0.08. The triangle residuals are -0.005 (012), 0.075 (013), -0.007 (023) and -0.087 (123). The two flagged triangles share exactly the edge 1–3, which RG identifies. This is case (c) of Theorem 2.2: no global object exists until the faulty conversion is repaired.

A3, control. At $\kappa = 0$ the cocycle closes and the class vanishes. The global map is accepted up to one additive null, the common zero.

4 Series B: different lenses, one geometry

A sphere of radius 2 carries 400 landmarks, with a shared triangulation of 1,194 edges and a physical rod equal to one edge. Three observers see it through different lenses.

Lens	χ	curvature $\times$ rod^2	raw rod reading
ambient chords, metres	2	0.03302	0.362
coarse geodesics, units of 1/3.7 m	2	0.03310	1.340
two-chart stereographic atlas with metric, cm	2	0.03295	36.22

The true value is rod$^2/R^2 = 0.03279$, i.e. $R = 5.52$ rods. The Euler characteristic is exact for every lens, by the combinatorial Gauss–Bonnet theorem: the sum of deficits is $2\pi\chi$ for *any* admissible edge lengths. The curvature in anchor units agrees to 0.5%, including the chord-versus-geodesic bias. Raw readings differ by a factor of 100.

Typing:

- *invariant:* χ, and curvature in anchor units;
- *representative:* coordinates and raw units;
- *lens-limited:* a one-dimensional observer measuring a single loop length sees $2\pi R$, but cannot identify curvature without the declaration "this loop is a great circle". Angle deficits need 2-cells.

Agreement here is structural: it survives a change of embedding, chart, unit and resolution.

5 Series C: what each observer forgets, and what the family remembers

Each of three observers masks a different polar cap and sees 91.25%, 91.5% and 91.5% of the landmarks. Jointly they see 100%: the joint fibre is the intersection of the individual fibres, and here it is trivial on the landmarks.

A cap of 24 landmarks hidden from all three is a permanent fibre for this family. A radial deformation inside it changes inside edge lengths by up to 0.072 and outside edge lengths by exactly 0. So every record of every observer is unchanged, while the geometry inside differs. RG reports this region as *hidden for the family*: it is recoverable only by a protocol that sees it. It is never silently filled.

6 Series D: orientation from redundancy

Here 200 observers have unknown headings and measure noisy relative headings ($\sigma = 0.3$) on a circulant graph, and RG synchronizes them by the spectral method [3].

degree	relative rms	spectral gap	decision
2	1.12	9×10^{-5}	abstain
4	1.11	0.0015	abstain
8	0.30	0.0033	abstain (conservative)
16	0.13	0.011	estimate
32	0.058	0.043	estimate

Findings.

- Redundancy orients: the error falls as the degree rises.
- The abstention rule, based on the spectral gap, is sound. It never estimates where the error exceeds 0.5, and it abstains conservatively at degree 8.
- The *global* rotation is null: without an anchor the absolute rms is 2.31, and with one observer's physical heading it is 0.14.
- Among 30 fully connected observers, a corrupted observer (number 17) is located by counting triangle residuals above 0.3.

7 Series E: the scale null and the physical anchor

Observers at scales 0.1, 1 and 10 read four lengths. The raw readings differ by a factor of 100, for example $0.070, \ldots, 0.50$ against $\ldots, 50$, while the ratios agree. Dividing by each observer's own reading of one physical rod gives 0.699, 1.298, 2.915 and 5.030, against the true 0.7, 1.3, 2.9 and 5.0. Absolute size is a null until a physical anchor is declared, as in the print's unit-anchor rule.

8 Series F: does RG fake structure?

Over 200 seeded trials for each condition, the overlap-gluing test accepts:

- genuine overlaps: 1.00;
- shuffled labels: 0.00;
- pure noise: 0.00;
- a reflected chart offered as a translation: 0.00.

RG does not manufacture agreement. Each fake fails the declared conversion class, and nothing is glued.

9 Series G: edges, overlaps and bad sampling

The uncertainty of a transition estimated from m overlap samples is 0.020, 0.014, 0.0093, 0.0052 and 0.0026 for $m = 1$, 2, 4, 16 and 64, scaling as $m^{-1/2}$. A single sample gives a value but no variance, so no certificate. RG abstains there.

Edge regions with thin overlaps therefore carry large conversion uncertainty, and this propagates into σ_δ and the decision rule. Covers must also be *good*. Two arcs covering a circle overlap in two components, and the resulting nerve gives $H^1 = 0$, which is wrong. Three arcs give $H^1 = 1$ (nerve theorem; Structural Closure).

10 What agreement establishes

Claim	Status	Evidence
Local reconstructions are valid	ESTABLISHED	pairwise residuals (A1); local readings retained
A global chart exists	ONLY IF $[g] = 0$	Theorem 2.2(b); A1 rejects it at $\kappa > 0$

Claim	Status	Evidence
A global object exists	ONLY IF cocycle	Theorem 2.2(c); A2 locates the violation
Invariants agree across lenses	ESTABLISHED	B: χ exact; curvature in rods within 0.5%
Raw coordinates agree	NOT (representative)	B: factor 100
Hidden-for-family regions	RETAINED as fibres	C: an exact alternative geometry
Global rotation and scale	NULL until anchored	D, E
Structure from noise or shuffles	REFUSED	F: 0/200
Thin overlaps and bad covers	ABSTAIN	G

11 Placement in the print, and limits

- **Print rungs.** Descent and quotients (L0–L19), rooms and the obstruction tower (L129–L131), atlas gluing (A6), mesh closure and coverage (Spectral Identifiability and Structural Closure), unit anchors (L94–L109), and the null and hidden typing (Engagement Case).
- **Battery.** `rg_agreement.py` runs Series A–G with seeded generators. Reruns are byte-identical.
- **Limits.**
  - All data are synthetic, with chosen models and noise.
  - The decision thresholds (3σ and 5σ, a spectral gap of 0.01, a gluing tolerance of 0.05) are declared, not derived.
  - The faking tests cover three named failure modes, not all adversaries.
  - The 1-D lens limitation is shown for one loop observer.
  - The descent theorem is the standard gluing and Čech result, imported.

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